Clinical Nutrition 40 (2021) 303–312

Contents lists available at ScienceDirect

Clinical Nutrition

journal homepage: http://www.elsevier.com/locate/clnu

Original article

Accuracy of surrogate methods to estimate skeletal muscle mass in non-dialysis dependent patients with chronic kidney disease and in kidney transplant recipients

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ARTICLE INFO

Article history:
Received 21 November 2019
Accepted 8 May 2020

Keywords:
Muscle mass
Accuracy
Chronic kidney disease
Kidney transplantation
Bioelectrical impedance analysis
Dual energy X-ray absorptiometry

SUMMARY

Background & aims: Bioelectrical impedance analysis (BIA) and anthropometric predictive equations have been proposed to estimate whole-body (SMM) and appendicular skeletal muscle mass (ASM) as surrogate for dual energy X-ray absorptiometry (DXA) in distinct population groups. However, their accuracy in estimating body composition in non-dialysis dependent patients with chronic kidney disease (NDD-CKD) and kidney transplant recipients (KTR) is unknown. The aim of this study was to investigate the accuracy and reproducibility of BIA and anthropometric predictive equations in estimating SMM and ASM compared to DXA, in NDD-CKD patients and KTR.

Methods: A cross-sectional study including adult NDD-CKD patients and KTR, with body mass index (BMI) $\geq$18.5 kg/m^2. ASM and estimated SMM were evaluated by DXA, BIA (Janssen, Kyle and MacDonald equations) and anthropometry (Lee and Baumgartner equations). Low muscle mass (LowMM) was defined according to cutoffs proposed by guidelines for ASM, ASM/height^2 and ASM/BMI. The best performing equation as surrogate for DXA, considering both groups of studied patients, was defined based in the highest Lin's concordance correlation coefficient (CCC) value, the lowest Bland–Altman bias (<1.5 kg) combined with the narrowest upper and lower limits of agreement (LoA), and the highest Cohen's kappa values for the low muscle mass diagnosis.

Results: Studied groups comprised NDD-CKD patients (n = 321: males = 55.1%; 65.4 ± 13.1 years; eGFR = 28.8 ± 12.7 ml/min) and KTR (n = 200: males = 57.7%; 47.5 ± 11.3 years; eGFR = 54.7 ± 20.7 ml/min). In both groups, the predictive equations presenting the best accuracy compared to DXA were SMM-BIA-Janssen (NDD-CKD patients: CCC = 0.88, 95%CI = 0.83–0.92; bias = 0.0 kg; KTR: CCC = 0.89, 95%CI = 0.86–0.92, bias = −1.2 kg) and ASM-BIA-Kyle (NDD-CKD patients: CCC = 0.87, 95%CI = 0.82–0.90, bias = 0.7 kg; KTR: CCC = 0.89, 95%CI = 0.86–0.92, bias = −0.8 kg). In NDD-CKD patients and KTR, LowMM frequency was similar according to ASM-BIA-Kyle versus ASM-DXA. The reproducibility and inter-agreement to diagnose LowMM using ASM/height^2 and ASM/BMI estimated by BIA-Kyle equation versus DXA was moderate (kappa: 0.41–0.60), in both groups. Whereas female patients showed higher inter-agreement (AUC>80%) when ASM/BMI index was used, male patients presented higher AUC (70–74%; slightly <80%) for ASM/height^2 index.

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https://doi.org/10.1016/j.clnu.2020.05.021
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Conclusions: The predictive equations with best performance to assess muscle mass in both NDD-CKD patients and KTR was SMM-BIA by Janssen and ASM-BIA by Kyle. The reproducibility to diagnose low muscle mass, comparing BIA with DXA, was high using ASM/BMI in females and ASM/height^2 in males in both groups.

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Abbreviations

SMM	Skeletal muscle mass
CKD	Chronic kidney disease
NDD-CKD	Non-dialysis dependent CKD
KTR	Kidney transplantation recipients
DXA	Dual-energy X-ray absorptiometry
LST	Lean soft tissue
ASM	Appendicular muscle mass
BIA	Bioelectrical impedance analysis
HGS	Hand grip strength
eGFR	Estimated glomerular filtration rate
HC	Hip circumference
BMI	Body mass index
TBW	Total body water
PBW-FFM	Percentage body water of the fat-free mass
CKD-EPI	Chronic Kidney Disease-Epidemiology Collaboration
EWGSOP2	European Working Group on Sarcopenia in Older People
FNIH	Foundation of the National Institutes of Health
SD	Standard deviation
IqI	Interquartile interval
CCC	Lin's concordance correlation coefficient
AUC	Area under the curve
CT	Computed tomography
MRI	Magnetic resonance imaging

1. Introduction

Whole-body skeletal muscle mass (SMM) reflects the majority of body protein mass and can be accurately measured using specialized equipment. The evaluation of SMM is important to nutritional status assessment and may be useful in treatment decisions [1,2]. Loss of SMM is multifactorial including disuse, altered endocrine function, nutritional deficiencies, inflammation, insulin resistance and chronic diseases, such as chronic kidney disease (CKD) [2,3].

CKD is a worldwide health problem with high rates of mortality [4]. A variety of conditions inherent to CKD, such as metabolic disorders (e.g.: acidosis and insulin resistance), chronic inflammation and decreased physical activity may negatively impact body composition leading to SMM loss [5–12]. Some of these metabolic alterations start in early stages of CKD and develop in parallel with deterioration of renal function, until the loss of kidney function, when dialysis therapy or kidney transplantation must be instituted [4,10,13]. Kidney transplantation improves renal function and the uremic conditions [13]. However, factors inherent to pre-transplant (e.g.: uremia throughout CKD progression, type and duration of dialysis) and also to post-transplant therapy (e.g.: immunosuppressive agents), may unfavorably impact body composition, resulting in low SMM even in the presence of high adiposity [14–16]. Patients with non-dialysis dependent CKD (NDD-CKD) and kidney transplantation recipients (KTR) presenting with low SMM are at higher risk for renal function decline, adverse cardiovascular events and mortality [17–20].

Albeit of limited availability, gold standard methods providing precise and reliable SMM measurements include computed tomography (CT) and magnetic resonance imaging (MRI) [1,2,21]. Likewise, dual-energy X-ray absorptiometry (DXA) is a high precision technique widely used for the assessment of total lean soft tissue (LST), and is highly correlated with gold standard methods to estimate whole-body SMM, a critical component of the LST compartment [2,21]. The amount of LST in the arms and legs, derived from DXA scans, is mainly muscle (excluding connective tissue and skin), providing the appendicular LST or appendicular muscle mass (ASM) [2,21,22]. Kim et al. (2002) validated a whole-body SMM prediction equation by using ASM from DXA [23]. DXA is among imaging techniques most commonly used in research settings to evaluate SMM in CKD patients under maintenance hemodialysis, as well as in NDD-CKD patients and KTR [7,11,17,24–29].

In addition to accuracy, the availability, cost and practicality of assessing SMM in routine clinical settings should be considered [1,21]. Bioelectrical impedance analysis (BIA) and anthropometry are surrogate methods for body composition estimation. Equations using BIA parameters have been validated to predict fat-free mass (FFM, LST + bone mineral content) and SMM in healthy individuals [30–34], as well as ASM in heart, lung and liver transplant patients [32]. Macdonald et al. (2006) [34] provided a validated equation to predict ASM in patients with CKD and observed a good precision in spite of patient's varied estimated glomerular filtration rate (eGFR: 14.3–133.2 ml/min/1.73 m^2). Anthropometric measures (weight and height) from healthy subjects have been used to predict SMM in an equation validated by Lee et al. (2000) [35]. ASM has also been estimated using anthropometry and hand grip strength (HGS) [36].

The above mentioned equations to estimate SMM and ASM have not yet been validated for NDD-CKD patients and KTR [37]. The routinely use of practical tools such as BIA and anthropometry in clinical settings may enhance the assessment of nutritional status, and therefore, the nutrition care process. Therefore, the primary aim of this study was to assess the accuracy and reproducibility of equations using BIA and anthropometric parameters, compared to DXA, to estimate SMM and ASM in NDD-CKD patients and KTR.

2. Methods

This cross-sectional study evaluated clinically stable NDD-CKD patients and KTR, under regular treatment at a nephrology interdisciplinary outpatient clinic at Pedro Ernesto University Hospital (Rio de Janeiro State University–Rio de Janeiro, Brazil). Patients were enrolled between 2015 and 2017. Inclusion criteria were: age ≥18 years, estimated glomerular filtration rate (eGFR) < 60 ml/min

(only for NDD-CKD patients), under regular treatment with a renal dietitian and nephrologist for at least 6 months (routine appointments scheduled for 4 to 6 visits per year, depending on clinical conditions and CKD stage).

Exclusion criteria were: presence of active malignant diseases, human immunodeficiency virus-HIV, autoimmune diseases, acute inflammation, chronic lung disease, liver failure and heart failure (class 3 or 4), patients undergoing dialysis and immunosuppressive drugs therapy (only for NDD-CKD patients), pregnant or lactating women, and patients with apparent edema or any conditions causing significant uncontrolled clinically apparent oedema. The protocol was approved by the ethical human research committee of Pedro Ernesto University Hospital, and all patients gave written consent before inclusion in the study.

2.1. Protocol and data collection

Demographic, clinical/laboratorial, anthropometric, and nutritional data were collected. Patients were advised to fast for 12 h prior the assessment day, avoid strenuous physical activity, and to follow their usual diet prescribed during routine visits. Anthropometric measurements and BIA were performed by two experienced and trained dietitians, whereas DXA procedure was performed by a trained technician, all evaluations on the same day.

2.2. Anthropometric measurements

Anthropometric measurements included body weight, height, and waist and hip circumferences. These were obtained with patients standing erect with their head in the Frankfort plane, wearing light clothing, no shoes and with empty bladder. Body weight and height were measured to the nearest 0.1 kg and 0.5 cm, respectively, using a balance beam scale Filizola® (São Paulo, Brazil). Hip circumference (HC) was measured using a flexible plastic tape with graduated scale at the widest point over the hip/buttocks area with the tape parallel to the floor [38]. Body mass index (BMI) was calculated for each patient as weight (kg)/height (m^2), and classified according to the World Health Organization as normal (18.5–24.9 kg/m^2) and as overweight/obese (≥25 kg/m^2) [39].

Muscle strength was assessed using HGS. Measurements were obtained using a handheld dynamometer (Baseline® Smedley Spring Dynamometer - Fabrication Enterprises Inc., NY, USA), according to the protocol recommended by the American Association of Hand Therapists [40]. Patients were first familiarized with the device, and then put in position (seated, shoulders adducted and neutrally rotated, elbow flexed at 90°, forearm in neutral and wrist between 0 and 30° of dorsiflexion), and instructed to grip the dynamometer with the maximum strength in response to a voice command. Three measurements were obtained by 1-min intervals for each hand in a rotational way, with the highest value of three measurements in each hand being considered for the study.

2.3. BIA

A tetrapolar single frequency BIA was used to perform the analysis (Model 310 generator and body; Biodynamics®, Seattle, WA). Patients were in a supine position with arms and legs lying parallel and apart. Two electrodes were placed on the hand and wrist, and two others were positioned on the foot and ankle of the right side of the body. An electrical current of 800 A at 50 kHz was then applied to the body, and resistance and reactance were measured. The BIA parameters evaluated were resistance, reactance, total body water (TBW), percentage body water of the fat-free mass (PBW-FFM), total body fat provided by the software manufacturer.

The hydration status of both of our cohorts was compared with a cohort with normal hydration status under monitoring and control of blood pressure followed by our group (without CKD and diabetes) as comparison group (n = 64; age = 56.8 ± 6.0 years; BMI = 28.3 ± 3.0 kg/m^2) [41] Thus, we explored the BIA-derived resistance index used in predictive equations, calculated as: RIndex = [height (Ht)^2/R; cm^2/ohms], TBW and PBW-FFM. The values were similar (ANOVA: $p > 0.05$) among three groups: a) RIndex (cm^2/ohms) in male patients (NDD-CKD patients = 58.5 ± 10.2; KTR = 64.1 ± 8.2; comparison group = 64.5 ± 10.6, p = 0.11), in female patients (NDD-CKD patients = 42.2 ± 4.6; KTR = 45.0 ± 8.2; comparison group = 44.0 ± 6.2, p = 0.53); b) TBW (liters) in male patients (NDD-CKD patients = 41.1 ± 6.0; KTR = 44.4 ± 5.1; comparison group = 46.0 ± 7.2, p = 0.07), in female patients (NDD-CKD patients = 31.9 ± 6.2; KTR = 30.6 ± 3.2; comparison group = 32.9 ± 4.3, p = 0.06); c) PBW-FFM (%)in male patients (NDD-CKD = 72.8 ± 1.5; KTR = 72.6 ± 1.1; comparison group = 74.7 ± 5.8, p = 0.15), in female patients (NDD-CKD = 74.7 ± 3.8; KTR = 73.3 ± 2.5; comparison group = 73.7 ± 3.2, p = 0.12). Studied patients were routinely monitored for blood pressure and treated to be maintained among recommended limits [4,42,43], no clinical signs of edema were reported in patients' medical records.

2.4. DXA

DXA was performed using a Lunar DPX Bone Densitometer scanner (iDXA, GE Healthcare Lunar Radiation®, Madison, WI - software enCore 2011 version 13.60). Fat tissue (grams), fat mass (%), LST (grams) and bone mineral mass (grams) were calculated. FFM was calculated as the sum of LST plus bone mineral mass.

2.5. Laboratory parameters

Patients underwent blood sampling early in the morning, after overnight fasting. Laboratory analyses were performed at the Central laboratory and at the Laboratory of Endocrine Physiology of Pedro Ernesto University Hospital, The following were assessed: urea and uric acid by urease kinetic urease method; potassium by selective electrode method; phosphorus by phosphomolybdate method; glucose by enzymatic hexokinase method by automated technique; hemoglobin by spectrophotometry; albumin was determined by colorimetric method; and creatinine by kinetic method calibrated to IDMS: COBAS 6000 (Roche/Hitachi). eGFR was estimated using the Chronic Kidney Disease-Epidemiology Collaboration (CKD-EPI) equation [4].

2.6. Body compartments and predictive equations of the study

SMM was estimated by different equations based on measures obtained by DXA, BIA and anthropometry as follow bellow.

2.6.1. SMM predictive equations

SMM-DXA$_{Kim}$ [23] = [1.13 **x** ASM (by DXA; kg)] **-** [0.02 x Age (years)] + [0.61 x Sex (men = 1, women = 0)] + 0.97.

SMM-BIA$_{Janssen}$ [30] = [0.401 x [height (cm)^2 ÷ Resistance (by BIA; ohms)]] + [3.825 x Sex (men = 1, women = 0)] + [- 0.071 x Age (years)] + 5.102

SMM-Anthropometry$_{Lee}$ [35] = [0.244 **x** Weight (kg)] + [7.8 x Height(m)] **-** [0.098 x age (years)] + [6.6 x Sex (men = 1, women = 0)] + [3.3 **x** ethnic factor (0 for Hispanic)]

SMM-DXA$_{Kim}$ was used as a reference method against which SMM-BIA$_{Janssen}$ and SMM-Anthropometry$_{Lee}$ were compared against.

ASM was obtained by the sum of the LST from the limbs. ASM was also estimated by BIA parameters according to Kyle et al. (2003) [32] and Macdonald et al. (2006) [34], as well as by anthropometric measures and HGS according to Baumgartner et al. (1998) [36] equation, as follow bellow.

2.6.2. ASM predictive equations

ASM-BIA$_{Kyle}$ [32] = **-**4.211 + (0.267 x Height^2 ÷ Resistance) + [(0.095 **x** Weight (kg)] + [1.909 x Sex (men = 1, women = 0)] + (**-** 0.012 x Age) + (0.058 **x** Reactance)

ASM-BIA$_{Macdonald}$ [34] = **-**11.626 + (0.292 **X** Height^2 ÷ Resistance) + (0.06983 **x** Reactance) + (0.08553 **x** Height) + [**-**2 .092 **x** Sex (men = 0, women = 1)] + (**-** 0.05 **x** Age)

ASM-Anthropometry$_{Baumgartner}$ [36] = [0.2487 **X** Weight (KG)] + [0.0483 **X** Height (cm)] **-** [0.1584 **X** Hip circumference (cm)] + [0.0732 **X** Handgrip strength by dinamometer (KG)] + [2.5843 **X** Sex (men = 1, women = 0)] + 5.8828

ASM by DXA (ASM-DXA) was used as a reference method against which ASM-BIA$_{Kyle,}$ ASM-BIA$_{Macdonald}$, and ASM-ANTHROPOMETRY$_{Baumgartner}$ were compared against.

In order to obtain ASM indices, ASM was divided by height squared (ASM/height^2) [1] and by body mass index (BMI) (ASM/BMI) [1,44].

The predictive equations hereby evaluated used different BIA equipment in the development and validation studies as follow: the Janssen's et al. [30]: 101B analyzer, RJL Systems, Detroit, MI/USA; Kyle's et al. [32]: Xitron 4000B, Xitron Technologies, Inc, San Diego, CA/USA; Macdonald's et al. [34]: Hydra ECF/ICF 4200, Xitron Technologies, San Diego, CA/USA), all of them using tetrapolar electrodes with an operating frequency of 50 kHz at 800 mA, resistance and reactance were measured by the BIA generator.

2.7. Low muscle mass definition

As no cutpoints to define low muscle mass are available for both our cohorts, we used definitions based on the aging literature. As such, the presence of low muscle mass was defined according to ASM cutoffs values proposed for older adults using the European Working Group on Sarcopenia in Older People (EWGSOP2) [1] (ASM: men<20 kg and women<15 kg) or the Foundation of the National Institutes of Health (FNIH) [44] (ASM: men<19.75 g and women<15.02 kg); low muscle mass was also defined by using ASM indices according the cutoffs proposed by EWGSOP2 [1,45] (ASM/height^2: men<7.0 kg/m^2; women< 5.5 kg/m^2) and by FNIH [44] (ASM/BMI: men<0.789 kg/m^2; women<0.512 kg/m^2).

2.8. Statistical analyses

The normality of continuous variables was tested by the Kolmogorov–Smirnov test, and data was shown as mean ± standard deviation (SD) or median and interquartile interval (IqI) according to normal or non-normal distribution, respectively. Comparison of continuous variables between NDD-CKD and KTR was performed by independent T-test or by Mann–Whitney test for values with normal or non-normal distribution, respectively. Categorical variables were compared by χ^2-test. The accuracy of SMM and ASM values estimated by BIA and by anthropometry using the above described predictive equations were compared with values obtained by DXA.

2.8.1. Accuracy and precision analysis by concordance correlation coefficient

Lin's concordance correlation coefficient (CCC) [46] was calculated to analyse agreement, (as a product of accuracy and precision) by comparing muscle mass parameters values obtained by DXA (ASM and estimated SMM) versus values from each predictive equations. CCC ranges from −1 to 1, with a perfect strength-of-agreement considered 1.

2.8.2. Bland-Altman plots

Bland-Altman plot analysis [47] was applied to illustrate the individual agreement between FFM, SMM and ASM values obtained by DXA *versus* values estimated by predictive equations above described (using BIA and anthropometric parameters). The visual comparison of differences between the two methods (Y-axis, i.e., method A - method B) was plotted against their mean difference (X-axis, i.e., mean value for the two methodologies: method A + method B/2), and the ranges of values for which two methods were most concordant/discordant are provided. The 95% limits of individual agreement (LoA) between the two methods were calculated as the mean difference between the two methods ± 1.96 SD.

The agreement of predictive equations compared with DXA were also evaluated by multiple regression analysis. The R^2 values (unadjusted, adjusted by sex and R^2-change) were estimated and the square root of the sum of the squared differences was calculated (represented by the standard error of the estimate-SEE). This measured how far the values estimated by the predictive equations plotted on the y-axis varied around the regression line, i. e., the degree of error or distance between the sets of data (DXA and predicted equations values).

2.8.3. Reproducibility

The inter-agreement between DXA and predictive equations was quantified by kappa values and classified according to degree of agreement beyond the expected by chance. The sensitivity (to detect true positive cases) and specificity values (to detect true negative cases) were also estimated, along with area under the curve (AUC) analysis to assign good reproducibility for AUC values > 80%, and moderate if kappa 0.41–0.60 [48], substantial if kappa values 0.61–0.80 [49].

The best performing equation to be considered as surrogate for DXA in both group of studied patients was defined based on the combination of the following three criteria: a) the highest Lin's concordance correlation coefficient (CCC) value; b) Bland–Altman bias <1.5 kg combined with the narrowest lower and upper limits of agreement; c) and the highest Cohen's kappa values for the low muscle mass diagnosis.

The CCC values and Bland-Altman plots were performed by MedCalc for Windows, version 9.2 (MedCalc Software, Mariakerke, Belgium). All other statistical analyses were performed on the Statistical Package for the Social Sciences (SPSS version 20.0, Illinois, USA).

3. Results

The majority of NDD-CKD patients (n = 321; males: 55.1%; 65.4 ± 13.1 years) presented with stages 3B and 4 CKD (3A: 13.0%, 3B: 34.5%, 4: 36.1% and 5: 16.4%; $p < 0.0001$) with mean eGFR = 28.8 ± 12.7 ml/min. KTR (n = 200; males: 57.7%; 47.5 ± 11.3 years) presented with a mean eGFR = 54.7 ± 20.7 ml/min; 36.5% (n = 73) presented with eGFR = 83.7 ± 10.4 ml/min, and 63.5% (n = 127) with eGFR = 46.1 ± 14.1 ml/min. Hypertension was the most frequent co-morbidity for both NDD-CKD (67.6%) and KTR (79.1%). Time after transplantation was 116.0 months (IqI: 33.5–175.0), before transplant the majority (85%) of patients were under hemodialysis therapy (median = 36.0 months; IqI: 15.0–78.0) or under peritoneal dialysis (median period = 12.0 months; IqI: 5.5–58.0). Findings on laboratory markers for both groups are presented in complementary material (Table C1). Mean values of serum albumin and glucose were within normal range in both patients groups.

Mean body composition values measured by DXA and or estimated by predictive equations are shown in Table 1. Overweight/obesity frequency was 58.3% in NDD-CKD patients and 52.2% in KTR. BMI mean values was similar between sex in both groups and >25 kg/m^2. As expected, body fat was higher and SMM was lower in women compared to men. In both groups, resistance index was higher for males compared to females, although reactance was similar. HGS mean values were lower in females compared to males

Table 1
Body composition parameters of non-dialysis dependent chronic kidney disease patients (n = 321) and kidney transplant recipients (n = 200), estimated by DXA and by different predictive equations.

Body composition parameters[a]	NDD-CKD Patients		KTR	
	Men	Women	Men	Women
BMI (kg/m^2)	26.3 ± 4.6	27.0 ± 4.9	25.7 ± 4.2	27.1 ± 5.7
Body fat (by DXA; %)	30.4 ± 7.5	40.0 ± 6.7[b]	28.9 ± 7.6	42.2 ± 7.0[b]
Body fat (by BIA; %)	25.9 ± 7.4	29.2 ± 7.7[b]	24.4 ± 6.1	35.7 ± 5.7[b]
Resistance (ohms)	498.2 ± 77.9	565.7 ± 87.7[b]	505.0 ± 75.3	618.4 ± 89.7[b]
Resistance index (cm^2/ohms)	58.2 ± 9.9	40.4 ± 7.5[b]	56.0 ± 11.1	44.3 ± 9.6[b]
Reactance (ohms)[c]	51.2 (42.5–61.8)	57.0 (48.1–67.8)	60.6 (51.6–72.2)	66.5 (56.9–79.3)
Total body water (by BIA) (L)	39.9 ± 7.4	32.1 ± 6.6[b]	40.5 ± 6.4	30.8 ± 5.2[b]
Body water of the FFM (BIA) (%)	77.3 ± 3.2	75.2 ± 6.4[b]	72.1 ± 1.8	73.8 ± 3.1[b]
Skeletal muscle mass (kg)				
DXA_{Kim}	26.14 ± 5.27	18.86 ± 3.45[b]	26.48 ± 4.27	18.11 ± 4.13[b]
$BIA_{Janssen}$	27.29 ± 4.82	18.33 ± 3.67[b]	28.80 ± 3.66	17.86 ± 3.18[b]
$Anthropometry_{Lee}$	27.75 ± 4.2	18.28 ± 3.76[b]	29.83 ± 3.85	20.40 ± 4.31[b]
Appendicular skeletal muscle mass (kg)				
DXA	22.95 ± 4.55	16.90 ± 3.17[b]	22.95 ± 3.66	15.98 ± 3.60[b]
BIA_{Kyle}	22.31 ± 4.29	16.18 ± 3.19[b]	23.77 ± 3.28	16.86 ± 3.68[b]
$BIA_{Macdonald}$	19.87 ± 4.07	13.38 ± 3.13[b]	22.38 ± 2.97	14.60 ± 3.62[b]
$Anthropometry_{Baumgartner}$	21.27 ± 3.47	14.81 ± 2.68[b]	21.96 ± 2.85	15.00 ± 2.74[b]

Abbreviations: NDD-CKD (non-dialysis dependent chronic kidney disease patients); KTR (kidney transplant recipients); BMI (body mass index); DXA (dual x-ray absorptiometry); BIA (Bioelectrical impedance analysis); IqI (interquartile interval).
[a] Values expressed as Mean ± standard deviation.
[b] T-test results were significantly ($p < 0.05$) for comparisons between men versus women.
[c] Values expressed as Median (IqI); Mann–Whitney test for comparison between male and female patients ($p > 0.05$).

Table 2
Concordance correlation coefficient and Bland–Altman bias (LoA) of skeletal muscle mass and appendicular muscle mass skeletal among different predictive equations compared with DXA values, in NND-CKD and RTR patients.

	Lin's Correlation coefficient		Bland Altman	
	Accuracy (CCC) (95% IC)		mean of differences (kg) (limits of agreement)	
	NDD-CKD	KTR	NDD-CKD	KTR
Skeletal muscle mass by DXA_{Kim}				
vs. **$BIA_{Janssen}$**	0.88 (0.83–0.92)[a]	0.89 (0.86–0.92)[a]	0.0 (-6.2; 6.2)	−1.2 (-6.4; 4.0)
vs. **$Anthropometry_{Lee}$**	0.88 (0.85–0.91)	0.84 (0.80–0.87)	−0.9 (-6.5; 4.8)	−3.0 (-7.2; 1.2)
Appendicular skeletal muscle mass by DXA				
vs. **BIA_{Kyle}**	0.89 (0.82–0.90)[a]	0.90 (0.87–0.93)[a]	0.70 (-4.4; 5.9)	−0.8 (-4.8; 3.2)
vs. **$BIA_{Macdonald}$**	0.69 (0.61–0.76)	0.82 (0.77–0.86)	3.4 (-2.4; 9.2)	1.0 (-4.6; 6.6)
vs. **$Anthropometry_{Baumgartner}$**	0.82 (0.75–0.87)	0.93 (0.91–0.95)	2.1 (-2.5; 6.8)	0.8 (-2.3; 3.9)

CCC= Precision (Pearson correlation coefficient) **x** Accuracy bias correction factor (Lin, 1989).
Abbreviations: CCC: concordance correlation coefficient; NDD-CKD (non-dialysis dependent chronic kidney disease); KTR (kidney transplant recipients); DXA (dual x-ray absorptiometry); BIA (Bioelectrical impedance analysis).
[a] Both groups (NDD-CKD and KTR) presented CCC ≥0.85 and lower limit of 95%CI ≥ 0.80.

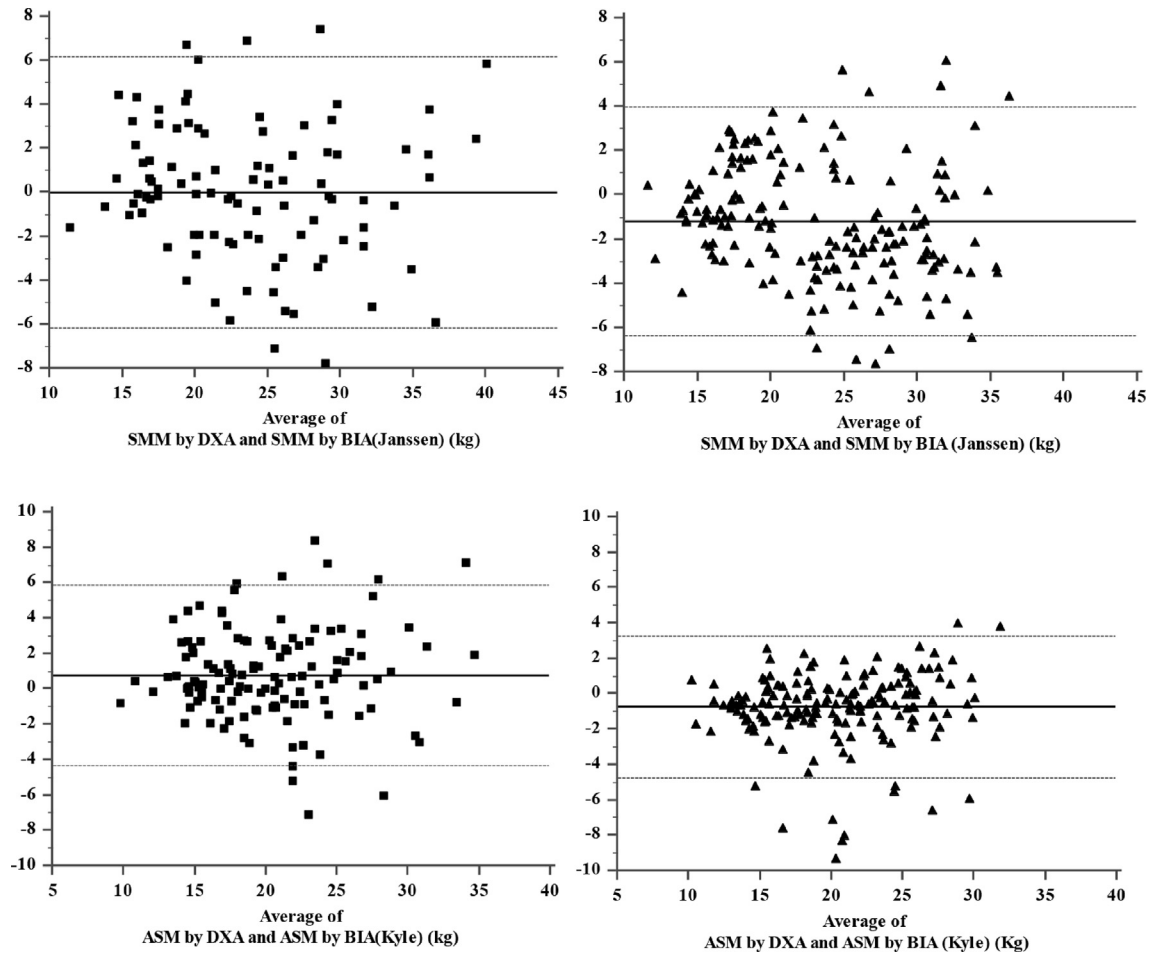

Fig. 1. Bland-Altman plots for SMM-BIA$_{Janssen}$ and for ASM-BIA$_{Kyle}$ predictive equations compared with DXA results, in NDD-CKD patients (■) and KTR (▲). legends: squares (■) are for NDD-CKD patients and triangles (▲) are for KTR. Abbreviations: NDD-CKD (non-dialysis dependent chronic kidney disease); KTR (kidney transplant receipts); SMM (skeletal muscle mass), ASM (appendicular muscle mass), DXA (Dual-energy X-ray absorptiometry); BIA (Bioelectrical impedance analysis).

Table 3
Unadjusted and sex adjusted regression analysis for the association between DXA and predictive equations, in NDD-CKD patients and KTR.

Regression Analysis[a]	Unadjusted		Adjusted by sex		R^2_{change}
	R^{2a}	SEE	R^{2a}	SEE	
NDD-CKD patients					
SMM-DXA$_{Kim}$ *vs.* SMM-BIA$_{Janssen}$	0.881	2.18	0.891	2.19	0.010
SMM-DXA$_{Kim}$ *vs.* SMM-BIA$_{Lee}$	0.798	2.62	0.807	2.56	0.010
ASM-DXA[b] *vs.* ASM-BIA$_{Kyle}$	0.865	2.16	0.863	2.17	0.002
ASM-DXA[b] *vs.* ASM-BIA$_{Macdonald}$	0.701	2.89	0.700	2.89	0.001
ASM-DXA[b] *vs.* ASM-BIA$_{Baumgartner}$	0.802	2.38	0.818	2.29	0.018
KTR					
SMM-DXA$_{Kim}$ *vs.* SMM-BIA$_{Janssen}$	0.833	2.43	0.840	2.37	0.007
SMM-DXA$_{Kim}$ *vs.* SMM-BIA$_{Lee}$	0.822	2.04	0.821	2.04	0.001
ASM-DXA[b] *vs.* ASM-BIA$_{Kyle}$	0.837	2.04	0.840	2.02	0.004
ASM-DXA[b] *vs.* ASM-BIA$_{Macdonald}$	0.700	2.77	0.705	2.74	0.005
ASM-DXA[b] *vs.* ASM-BIA$_{Baumgartner}$	0.827	1.60	0.827	1.60	0.000

Abbreviations: NDD-CKD (non-dialysis dependent chronic kidney disease); KTR (kidney transplant receipts); SMM (skeletal muscle mass), ASM (appendicular muscle mass), DXA (Dual-energy X-ray absorptiometry); BIA (Bioelectrical impedance analysis); SEE (standard error of the estimate: square root of the sum of the squared differences).

[a] Both (unadjusted and adjusted regression models) showed significant R2 ($p < 0.0001$).

[b] ($\sum$ ASM measured in 4 Limbs).

with NDD-CKD (females = 17.38 ± 7.77 and males = 28.15 ± 10.59 kg; $p < 0.0001$) and in KTR (females = 19.56 ± 5.56 and males = 34.61 ± 9.97 kg; $p < 0.0001$).

3.1. Accuracy and precision analysis by concordance correlation coefficient and Bland-Altman plots

The agreement, as a product of accuracy and precision, of estimated muscle mass (evaluated by predictive equations) in comparison to DXA is presented in Table 2. The Bland-Altman plots for SMM-BIA$_{Janssen}$ and for ASM-BIA$_{Kyle}$ equation compared with measured ASM-DXA are presented in Fig. 1. Bland-Altman plots for SMM-Anthropometry$_{Lee}$, ASM-Anthropometry$_{Baumgartner}$ and ASM-BIA$_{Macdonald}$ equations compared with DXA are presented in complementary material (Figure C1). Considering both groups (Table 2), the best performance according to the highest CCC values was observed for SMM-BIA$_{Janssen}$ and by ASM-BIA$_{Kyle}$ predictive equations, presenting with the lowest mean difference (<1.5 kg) compared to DXA. The CCC and lower limit of 95% was observed for SMM-Anthropometry$_{Lee}$ equation only in NDD-CKD patients, while for ASM-Anthropometry$_{Baumgartner}$ equation it was observed only in KTR patients. The CCC values for ASM-BIA$_{Macdonald}$ equation were lower in both groups of patients.

The agreement of the predictive equations was analysed according to the results of changes in R^2 and SEE (unadjusted and

Table 4
Reproducibility of BIA to evaluate low muscle mass compared to DXA as reference, in non-dialyzes patients with chronic kidney disease and in kidney transplant recipients, according to sex.

	Inter- agreement	Low muscle mass diagnostic performance			
	Kappa (p)	Sensibility (%)	Specificity (%)	AUC (95%CI)	Significance level P[b]
NDD-CKD patients					
[a]ASM/height^2					
DXA vs. ASM-BIA$_{Kyle}$					
All	0.40 (<0.0001)	64.71	85.05	0.75 (0.66–0.82)	0.0005
Men	**0.45 (< 0.0001)**	**55.56**	**91.94**	**0.74 (0.62–0.84)**	**0.017**
Women	0.35 (0.005)	75.00	75.6	0.75 (0.62–0.86)	0.0160
[a]ASM/BMI					
DXA vs. ASM-BIA$_{Kyle}$					
All	0.43 (<0.0001)	60.00	87.5	0.74 (0.65–0.81)	0.0004
Men	0.41 (0.001)	56.20	85.50	0.71 (0.59–0.81)	0.0085
Women	**0.44 (0.001)**	**75.00**	**89.80**	**0.82 (0.70–0.92)**	**0.0133**
KTR					
[a]ASM/height^2					
DXA vs. ASM-BIA$_{Kyle}$					
All	0.57 (0.0001)	48.94	99.1	0.74 (0.67–0.81)	<0.0001
Men	**0.51 (0.0001)**	**38.89**	**99.98**	**0.69 (0.59–0.78)**	**0.010**
Women	0.58 (0.0001)	55.17	97.92	0.77 (0.66–0.85)	0.000
[a]ASM/BMI					
DXA vs. ASM-BIA$_{Kyle}$					
All	0.56 (0.0001)	47.06	98.58	0.73 (0.66–0.79)	0.0001
Men	0.43 (0.0001)	35.00	98.72	0.67 (0.57–0.76)	0.000
Women	**0.71 (0.0001)**	**64.29**	**98.41**	**0.81 (0.71–0.89)**	**0.020**

Abbreviations: EWGSOP2 (European Working Group on Sarcopenia in Older People); FNIH (Foundation for the National Institutes of Health Sarcopenia Project); ASM (appendicular muscle mass); BMI (body mass index); DXA (Dual-energy X-ray absorptiometry); BIA (Bioelectrical impedance analysis).
[a] Patients were classified with low muscle mass according to cutoffs values for ASM/height^2 proposed by EWGSOP2 [1] (men< 7.0 kg; women<5.5 kg) and according to FNIH [44] cutoffs values for ASM/BMI (men<0.789 kg; women<0.512 kg).
[b] Asymptotic Significance: Null hypothesis for true area = 0.5.

adjusted by sex) (Table 3). The values of R^2 and SEE were not different after adjustment by sex. Significant regression models ($p < 0.0001$) were obtained. The variable sex showed low contribution to explain the variance between DXA and predictive equations values (accounting for only 0.0–1.8% of the variance). The adjusted R^2 was higher and SEE was lower for SMM-BIA$_{Janssen}$ and ASM-BIA$_{Kyle}$ equations compared to the others evaluated predictive equations.

3.2. Reproducibility

The frequency of patients with low muscle mass assessed by ASM-DXA and by ASM-BIA$_{Kyle}$, were similar ($p > 0.05$) in NDD-CKD patients (EWGSOP2: ASM-DXA = 22.6% *vs.* ASM-BIA$_{Kyle}$ = 31.5%; FNIH: ASM-DXA = 22.6% *vs.* ASM-BIA$_{Kyle}$ = 30.6%), and in KTR (EWGSOP2: ASM-DXA = 30.2% *vs.* ASM-BIA$_{Kyle=}$22.5%; FNIH: DXA = 29.7% *vs.* BIA$_{Kyle}$ = 21.4%).

The reproducibility of ASM-BIA$_{Kyle}$ equation in comparison with ASM-DXA was evaluated according to low muscle mass indices, using ASM/height^2 and by ASM/BMI, according the cutoffs proposed by EWGSOP2 and by FNIH. The inter-agreement was quantified by kappa values and according to sex (Table 4). In both groups of patients, the reproducibility of the indices (ASM/height^2 and by ASM/BMI) using ASM-BIA$_{Kyle}$ equation compared to ASM-DXA varied from moderate to substantial agreement (men and women kappa values were within 0.41–0.71, except for the unique kappa value of 0.35 observed in women of the NDD-CKD patients). In female patients from both groups (NDD-CKD and KTR), the higher kappa coefficients, sensitivity and specificity were obtained using ASM/BMI versus ASM/height^2 (ASM-BIA$_{Kyle}$), and the AUC value was high (>80%) indicating good reproducibility compared with measured ASM-DXA. In males from both groups, kappa coefficients, sensitivity and specificity values were higher when ASM-BIA$_{Kyle}$ index was ASM/height^2, presenting AUC (70–74%) slightly below the recommended value of 80%.

4. Discussion

The accuracy and reproducibility of BIA and anthropometric prediction equations to estimate ASM and SMM, in NDD-CKD patients and in KTR were analysed. The best performance was observed for SMM-BIA$_{Janssen}$ and by ASM-BIA$_{Kyle}$ equations, showing higher CCC values compared to SMM-DXA$_{Kim}$ and ASM-DXA. In females, the AUC showed good reproducibility for ASM/BMI, while in males the AUC values were acceptable for ASM/height^2.

The vast majority of studies evaluating the impact of muscle integrity on poor outcomes and mortality were conducted in patients with end-stage CKD and under dialysis therapy. Both these conditions are associated with impaired energy and protein resources which result in substantial muscle mass and function deterioration [50–53]. Data from NDD-CKD is limited; one study found that low SMM (assessed by BIA) and low HGS were associated with early systemic indices of atherosclerosis and endothelial dysfunction [54]; another study showed association between low SMM with mortality [20]. Low HGS has been associated with predialysis mortality or dialysis-dependent end-stage renal disease [55], cardiovascular hospitalization [56], and all-cause mortality [54,56]. Two longitudinal studies conducted in KTR (before-after transplantation) reported a decrease in FFM assessed by DXA [57] and by BIA [58], and one study reported a decrease in muscle mass assessed by an impedance analyzer [16]. No studies have been conducted on the impact of low SMM in KTR.

No method and cutpoint has been stablished as an absolute definition of low muscle mass, specially one considering sex, age, ethnicity and pathological conditions [1,3,44,59]. Reference techniques, such as CT, MRI and DXA are unlikely to be available and feasible in all clinical practices, whereas predictive equations using BIA and anthropometric measurements are the most widely reported surrogates methods. Advantages of BIA include safety, practicality, simplicity and general patient acceptance. However,

disease stage and clinical status, need to be taken into account. Raw BIA measurements (i.e. resistance and reactance) are proportional to body hydroelectrolytic status and are used to estimate TBW, which is used to compute FFM [60,61]. The reliability of BIA estimates is related with fluid overload, a condition that may occur in dialysis-dependent CKD patients [62–66]. In the present study both groups of patients were under regular treatment, and RIndex, TBW and PBW-FFM were similar to that of the normal hydration comparison group [41]. The RIndex is the best predictor of body fluid status [67] and is a component of SMM-BIA$_{Janssen}$ and ASM-BIA$_{Kyle}$ predictive equations used in our CKD cohort.

The predictive equation developed by Kyle et al. [31] to estimate ASM was strongly correlated with ASM-DXA as a reference method. This equation was used to evaluate ASM in patients with stages 1–5 CKD, showing bias = 2.3 kg (±4.9 kg) compared to ASM measured by DXA [34]. The accuracy of ASM-BIA$_{Kyle}$ equation in our studied population (both CKD groups) was good (CCC≥0.85) with bias <1.0 kg compared to ASM-DXA. This high accuracy may be due to the wide age range and different ethnicity of the population used in Kyle et al. [32] study (22–94 years, n = 444 healthy individuals) and the use of non-healthy individuals in the validation analysis (n = 326 pre- or post-heart, lung or liver transplant patients, 18–70 years). This would contribute to the equation's predictive power to be used for other populations, such as our CKD patients. Macdonald et al. (2006) [34] proposed a predictive equation to estimate ASM in patients with CKD stages 1–5 (n = 75; 66.7% men; 76.4% in stages 3 and 4; BMI = 27.9 ± 4.4 kg/m^2, body fat = 36.0 ± 9.6%, ASM/BIA = 21.0 ± 4.6 kg; hydration of the FFM = 75.9 ± 8.9%). In our KTR population the bias of ASM-BIA$_{Macdonald}$ equation compared to ASM-DXA was 1.0 kg, and in NDD-CKD patients it was 3.4 kg. The mean eGFR of patients in the Macdonald's study [34] (45.9 ± 28.8 ml/min) was similar to that of KTR (54.66 ± 20.62 ml/min), both at 3A stage CKD, but was higher than our NDD-CKD patients (28.81 ± 12.71; stage 4 CKD), despite similar age (Macdonald's study: 65.1 ± 12.0 years; NDD-CKD patients: 65.39 ± 13.07 years). In another study, Macdonald et al. [68] evaluated muscle mass in CKD patients and showed that age, sex, height and weight did not account for ASM variance, while serum creatinine levels did. Furthermore, ASM was strongly associated with eGFR. Thus, eGFR is an important variable to account for when using Macdonald's et al. equation [34] to estimate ASM.

We also explored SMM values, which large proportion is comprised of ASM [23]. The accuracy of SMM-BIA$_{Janssen}$ and SMM-Anthropometry$_{Lee}$ predictive equations [30,35] were compared with the SMM-DXA$_{Kim}$ validated equation [23]. The SMM-DXA$_{Kim}$ equation was developed (n = 321) and validated (n = 93) in an ethnically diverse group of healthy adults. Its accuracy was assessed by comparing SMM quantified by MRI using the same equipment used in SMM-BIA$_{Janssen}$ and SMM-Anthropometry$_{Lee}$. In the present study, SMM-Anthropometry$_{Lee}$ [35] showed good accuracy in NDD-CKD patients and KTR, but the underestimation in KTR was of 3.0 kg (95%CI: −7.2; 1.2). Conversely, the accuracy of SMM-BIA$_{Janssen}$ predictive equation was good and mean differences were <1.2 kg in both studied groups. The high accuracy of SMM-BIA$_{Janssen}$ observed for both groups may be related to the population characteristics, which comprised 388 healthy adults (59.3% men) in a wide age (18–86 years) and BMI (16–48 kg/m^2) range, and of varied ethnicity (caucasian, african-american, asian and hispanic). The rationale for exploring the accuracy of the Lee's et al. [35] equation (SMM-Anthropometry$_{Lee}$) was based on the feasibility of obtaining weight and height measures in clinical practice; besides, its validation included individuals, whose body composition resemble our studied population. The inconsistency of SMM-Anthropometry$_{Lee}$ [35] equation's performance in the present study may be related to the inclusion of healthy subjects, with normal BMI and obesity (but not overweight). The strong association between anthropometrics and estimated SMM must considered when using predictive equations based on anthropometry. Additional validation and accuracy studies are needed to test the usefulness of predictive equations in estimating SMM using anthropometric measures, contributing to the future development of population-specific equations.

Criteria (cutoff values) of LoA capable of diagnosing low or high muscle mass has not been established. No studies have reported the relationship between skeletal muscle mass (SMM and ASM: values and magnitudes of changes) with outcomes in NDD-CKD or KTR population. Most often, field techniques such as BIA are the only available to practitioners. As such, understanding the errors/limitations of BIA use for these populations is important. Our aim was to describe the performance of different surrogate methods (compared with DXA), and to provide SMM and ASM values that may contribute with muscle assessment in NDD-CKD and KTR patients. These equations should be used during clinical monitoring of patients throughout the nutrition care process. The use of a more practical technique during routine clinical care fulfill this gap and allows for future understanding of the applicability of this technique to monitor body composition in these patients.

The accuracy of low muscle mass diagnosis using ASM estimated by the validated BIA equation was analysed in comparison with DXA, following the worldwide expert consensus recommendations to define low muscle mass, including older adults [1,3,41,59]. The frequency of patients with low muscle mass assessed by ASM-DXA and by ASM-BIA$_{Kyle}$ was similar, the observed frequencies were higher using BIA compared to DXA for NDD-CKD patients and lower for KTR patients. Thus, a reproducibility analysis was performed. The best reproducibility in males was observed by ASM/height^2, while in females by ASM/BMI, suggesting that muscle mass in males is related to height and in women with high adiposity. Overall kappa values were moderate, indicating higher specificity than sensitivity. Thus, the used cutoff values may be limited to detect true positive cases of low muscle mass in the CKD population. Although the EWGSOP2 and FNIH cutpoints are not specific to patients with chronic diseases such as ours, a benchmark for estimating the prevalence of low muscle mass in these cohorts is provided. Further studies exploring cutpoints to define low muscle mass in CKD and KTR populations, and their comparability with the ones we hereby explored are needed.

In conclusion, accuracy and reproducibility analysis in the present study highlighted that BIA can be an alternative technique to assess muscle mass in CKD patients. The accurate results for ASM may contribute with an unified parameter enabling comparisons among different CKD populations. The accurate assessment of muscle mass in CKD population is a key aspect in identifying those at risk for muscle disorders, our findings show that BIA can be a clinically useful technique to define low muscle in this population, based on the ASM index estimates. The assessment of muscle mass in CKD patients can further our understanding on the impact of low muscle mass in clinical outcomes, prevention and management pathways, and to determine the potential of benefit from nutritional interventions.

The novelties of the present study include the lack of studies reporting crude values for SMM and ASM in NDD-CKD patients and KTR, which is important for future comparison within these specific populations. Additionally, our results highlight the value of using BIA as a surrogate clinical setting to assess and monitoring nutritional status, mainly body somatic protein compartment. Finally, we also report a discriminative analysis providing indices with higher sensitivity and specificity values with discriminative power to detect low muscle mass. The study's main limitation is the cross-sectional design which precludes an exposure/outcome association and the lack of a control health group, matched by age. Likewise,

the predictive equations hereby evaluated were developed and validated using different BIA equipment (see in Methods). Use of different machines impact data acquisition and equation development and should be interpret with caution as explained elsewhere [69]. Additionally, some limitations related to the use of DXA for body composition assessment should be considered such as different accuracy of lean mass computation among body regions (e.g. arms, legs, and thorax as in bone-free areas). Excess fat mass, and extreme hydration alterations and electrolyte status may also influence the result. Additionally, differences within and between manufacturers and software versions measurements should be considered when applicable, although this can be minimized by cross-calibration developed by cross-system comparison studies conducted with machines of different makes and models [2].

Funding

This work was supported by (1) Fundação de Amparo à Pesquisa do Estado do Rio de Janeiro [FAPERJ], Rio de Janeiro, Brazil and (2) Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES)

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Declaration of Competing Interest

None declared.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnu.2020.05.021.

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